

线性代数 (A) 2023-2024 学年秋季学期答案

timetraveler314

<https://chernoff.bond>

timetraveler314@outlook.com

1.

Solution. 注意到原矩阵实对称, 一定可以正交对角化. 只要求其特征向量并用 Gram-Schmidt 正交化即可. □

2.

Solution. 由最小二乘法和正交投影相关性质, 只需令 β 等于 α 在 U 中的投影, 此时 $\beta - \alpha$ 垂直于整个 U , 由勾股定理可以知道这就是使两者距离最小的 β .

这题算出来好像 α 落在 U 内... 不要怀疑自己. □

3.

Solution. (1) 奇异值即 $A^T A$ 的特征值的平方根.

(2) 右奇异向量 $v_1, \dots, v_n$ 即 $A^T A$ 的特征向量. 左奇异向量 $u_i = A \frac{v_i}{\sigma_i}$. 实际上我去年没有具体算出来左奇异向量, 只是列了式子, 也没有减分. □

4.

Solution. (1) 略.

(2) 类似(1), 只需要正交对角化即可.

(3) 这是 Rayleigh 商的经典应用, 答案是二次型矩阵最大的特征值. 可以参考这个讲义: <https://www.sjsu.edu/faculty/guangliang.chen/Math253S20/lec4RayleighQuotient.pdf>

原理即正交替换到新的坐标系后 (坐标变换 $y = Px \Rightarrow f(x) = y^T D y$). 正交变换保证模长 $\|x\| = \|y\| = 1$ 不变, 只要求在 $y_1^2 + \dots + y_n^2 = 1$ 条件下

$$f(y) = y_1^2 \lambda_1 + \dots + y_n^2 \lambda_n$$

的最大值即可.

不妨设 $\lambda_1 \geq \dots \geq \lambda_n$, 则容易看出 $f(y) \leq \lambda_1$, 且 $f(y) = \lambda_1$ 当且仅当 $y = e_1$. □

5.

Solution. 这是可对角化的一个重要性质. 矩阵 (即线性变换) 可对角化当且仅当全空间等于其特征空间的直和. 也就是有完备的特征向量组成的基. 这给了我们几何和代数上想象的空间. 实际上, 我们有如下的一般结论:

Theorem 5.1

设 n 阶矩阵 A 适合首一多项式 $g(x)$, 则 A 可对角化当且仅当 $g(x)$ 在 $\mathbb{C}$ 上无重根.

Proof. 设 $g(x) = (x - a_1)(x - a_2) \dots (x - a_n)$ 是 $\mathbb{C}$ 上的因式分解. 我们来证明:

$$\mathbb{C}^n = \text{Ker}(A - a_1 I) \oplus \dots \oplus \text{Ker}(A - a_n I)$$

□

设 $g_i(x) = \prod_{j \neq i} (x - a_j)$, 则 $\gcd(g_1, \dots, g_n) = 1$, 由 Bézout 定理, 存在 $h_1, \dots, h_n$ 使得 $h_1 g_1 + \dots + h_n g_n = 1$.

代入 $x = A$, 可得恒等式

$$g_1(A)h_1(A) + \dots + g_n(A)h_n(A) = I$$

对任意 $\alpha \in \mathbb{C}^n$, 有

$$\alpha = g_1(A)h_1(A)\alpha + \dots + g_n(A)h_n(A)\alpha$$

注意到: $(A - a_i I)g_i(A)h_i(A)\alpha = 0$, 所以 $g_i(A)h_i(A)\alpha \in \text{Ker}(A - a_i I)$. 于是由线性空间和的意义, 我们有

$$\mathbb{C}^n = \text{Ker}(A - a_1 I) + \dots + \text{Ker}(A - a_n I)$$

要证明上式是直和, 只需证明:

任取 $\alpha \in \text{Ker}(A - a_1 I) \cap (\text{Ker}(A - a_2 I) + \dots + \text{Ker}(A - a_n I))$, 则 $\alpha = \alpha_2 + \dots + \alpha_n$, 其中 $\alpha_i \in \text{Ker}(A - a_i I)$.

可知

$$\alpha = h_1(A)g_1(A)(\alpha_2 + \dots + \alpha_n) + h_2(A)g_2(A)\alpha_2 + \dots + h_n(A)g_n(A)\alpha_n = 0.$$

注意到上面推导的下标可任意选, 因此我们得到了直和的结论.

Note 5.1

实际上, 这是特征空间的直和分解, 学到 Jordan 标准型以及准素分解等内容后, 我们会发现这个结论的深刻意义.

回到原题, 利用上述定理, 我们看到 $g(A) = A^2 + I$ 的根是 i 和 $-i$, 因此 A 可对角化. 且 A 的所有特征值一定都是 i 和 $-i$ 之一, 因此与 A 相似的矩阵一定是 $\text{diag}(i, \dots, i, -i, \dots, -i)$ 的形式. □

6.

这是著名的降秩定理, 其中用到的分块矩阵技巧在这类特征值问题中非常好用, 以下从我的笔记中摘取相关定理:

Sylvester's Theorem is a fantastic result using the technique of blocked matrices. The idea can be tweaked and applied to many other problems. Here are some examples.

Theorem 6.1 The First Rank Reduction Theorem

Let A be invertible, then

$$\text{rank} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \text{rank } A + \text{rank}(D - CA^{-1}B).$$

Proof. Apply elementary blocked matrix operations to the matrix $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$:

$$\begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} I & -A^{-1}B \\ 0 & I \end{pmatrix} = \begin{pmatrix} A & 0 \\ 0 & D - CA^{-1}B \end{pmatrix}.$$

The result follows. □

Theorem 6.2 The Second Rank Reduction Theorem

Let A, D be invertible, then

$$\text{rank } A + \text{rank}(D - CA^{-1}B) = \text{rank } D + \text{rank}(A - BD^{-1}C).$$

Proof. Similar to the first theorem. □

Theorem 6.3 Sylvester's Inequality

$$\text{rank } AB \geq \text{rank } A + \text{rank } B - n.$$

Proof. Apply the first rank reduction theorem, or use the trick on

$$\begin{pmatrix} A & 0 \\ I & B \end{pmatrix} \rightarrow \begin{pmatrix} 0 & AB \\ I & 0 \end{pmatrix}.$$

The result follows from

$$\text{rank} \begin{pmatrix} A & 0 \\ I & B \end{pmatrix} \geq \text{rank } A + \text{rank } B.$$

Note 6.1

This is a direct result of the Frobenius Inequality. □

Theorem 6.4 Frobenius Inequality

$$\text{rank } ABC \geq \text{rank } AB + \text{rank } BC - \text{rank } B.$$

Proof.

$$\begin{pmatrix} ABC & 0 \\ 0 & B \end{pmatrix} \rightarrow \begin{pmatrix} ABC & AB \\ 0 & B \end{pmatrix} \rightarrow \begin{pmatrix} 0 & AB \\ -BC & B \end{pmatrix}.$$

□

As for eigenvalues and eigenvectors, we want to obtain the form $\lambda I - AB$, and this can be done similarly.

Theorem 6.5 Sylvester's Theorem for Eigenvalues

Let A be an $m \times n$ matrix, B be an $n \times m$ matrix, and $m \geq n$. We have

$$|\lambda I_m - AB| = \lambda^{m-n} |\lambda I_n - BA|.$$

Proof. A direct result of the first theorem when $\lambda \neq 0$.

When $\lambda = 0$, it can either be done by the Cauchy-Binet formula or by the perbutation method. □

The theorem reveals that AB and BA have the same non-zero eigenvalues with the same algebraic multiplicities. Furthermore, we can actually show that their geometric multiplicities are also the same.

Corollary 6.1

Let A be an $m \times n$ matrix, B be an $n \times m$ matrix. Then AB and BA have the same non-zero eigenvalues with the same algebraic and geometric multiplicities.

Proof. First theorem with

$$\begin{pmatrix} \lambda I_m & 0 \\ B & \lambda I_n - BA \end{pmatrix} \leftarrow \begin{pmatrix} \lambda I_m & A \\ B & I_n \end{pmatrix} \rightarrow \begin{pmatrix} \lambda I_m - AB & A \\ 0 & I_n \end{pmatrix}.$$

$$n + (n - \dim \ker(\lambda I_n - BA)) = n + (n - \dim \ker(\lambda I_m - AB)).$$

□

Proof. 原题两问的结论即 Corollary 6.1. □

7.

Solution. Linear Algebra Done Right 的习题. 笔者证明有误, 喜提 -8 分.

14. Solution: Since $e_1, \dots, e_n$ is an orthonormal basis of V , we have $\dim V = n$. To show that $v_1, \dots, v_n$ is a basis of V , it suffices to show that $v_1, \dots, v_n$ is linearly independent. We prove it by contradiction.

Suppose $v_1, \dots, v_n$ is linearly dependent, then there exist $a_1, \dots, a_n \in \mathbb{F}$ such that $a_k \neq 0$ for some $k \in \{1, \dots, n\}$ and

$$\sum_{i=1}^n a_i v_i = 0.$$

On one hand, by 6.25, we have

$$\left\| \sum_{i=1}^n a_i (e_i - v_i) \right\|^2 = \left\| \sum_{i=1}^n a_i e_i \right\|^2 = \sum_{i=1}^n |a_i|^2.$$

On the other hand, we also have

$$\begin{aligned} \left\| \sum_{i=1}^n a_i (e_i - v_i) \right\|^2 &= \left\langle \sum_{i=1}^n a_i (e_i - v_i), \sum_{j=1}^n a_j (e_j - v_j) \right\rangle \\ &= \sum_{i=1}^n \sum_{j=1}^n \left\langle a_i (e_i - v_i), a_j (e_j - v_j) \right\rangle \\ &\leq \left| \sum_{i=1}^n \sum_{j=1}^n \left\langle a_i (e_i - v_i), a_j (e_j - v_j) \right\rangle \right| \\ &\leq \sum_{i=1}^n \sum_{j=1}^n |\langle a_i (e_i - v_i), a_j (e_j - v_j) \rangle| \\ &\stackrel{\text{by 6.15}}{\leq} \sum_{i=1}^n \sum_{j=1}^n \|a_i (e_i - v_i)\| \|a_j (e_j - v_j)\| \\ &= \sum_{i=1}^n \sum_{j=1}^n |a_i| |a_j| \|e_i - v_i\| \|e_j - v_j\| \\ &\stackrel{\text{by assumption and } a_k \neq 0}{<} \sum_{i=1}^n \sum_{j=1}^n \frac{1}{n} |a_i| |a_j| = \frac{1}{n} \left(\sum_{i=1}^n |a_i| \right)^2 \\ &\stackrel{\text{by Problem 6.A.12}}{\leq} \sum_{i=1}^n |a_i|^2. \end{aligned}$$

Hence we get

$$\sum_{i=1}^n |a_i|^2 < \sum_{i=1}^n |a_i|^2,$$

which is impossible, hence completing the proof.

□

8.

Question 8.1

$f(x) = x^T A x$, A is a Gram matrix (or a positive-definite matrix), then

$$g(x) = \det \begin{pmatrix} A & x \\ x^T & 0 \end{pmatrix}$$

is negative definite.

Proof. #1 : Directly calculate the determinant. We have the lemma below:

Lemma 8.1

$$\begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} & x_1 \\ a_{21} & a_{22} & \dots & a_{2n} & x_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} & x_n \\ y_1 & y_2 & \dots & y_n & z \end{vmatrix} = z|A| - \sum \sum A_{ij} x_i y_j.$$

Proof. #1 (Complicated) : Expand the determinant by the last row, the first term is

$$(-1)^{n+2} x_1 \begin{vmatrix} a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \\ y_1 & y_2 & \dots & y_n \end{vmatrix}.$$

Expand the determinant by the last column, we have

$$(-1)^{n+2} x_1 (-1)^{n+1} (y_{11} A_{11} + y_{12} A_{12} + \dots + y_{1n} A_{1n}) = - \sum x_1 A_{1j} y_j.$$

Similarly, the i -th term is

$$- \sum x_i A_{ij} y_j.$$

Hence the lemma is proved. □

This problem gives that $g(x) = -x^T A x < 0$.

#2(Elementary) :

做分块初等变换 (好用, 好用, 好用):

$$\begin{pmatrix} A & x \\ x^T & 0 \end{pmatrix} \rightarrow \begin{pmatrix} A & x \\ 0 & -x^T A^{-1} x \end{pmatrix}$$

$$g(x) = \underbrace{\det A}_{>0} \cdot (-x^T A^{-1} x) < 0.$$

□